

COMPOSITE MATERIAL DATASHEETS CORE NOMEX OVER-EXP. HONEYCOMB

MATERIAL IDENTIFICATION

EU 1E001 - This data is controlled under EU Dual Use Regulation 2021/821, as amended. Authorization may be required for the export of the data outside of the EU.

Short informal name	NOMEX-OX
Commercial name	Hexcel HRH 10/OX, Eurocomposites ECA-R, Showa SAH-OX, etc.
Specifications	AIMS11-01-005 (Grades A, F & H)
Description	Honeycomb Cores, Nomex Polyamide Paper (meta-aramid), Phenolic Resin Impregnated, Over-expanded.
Version	1.0

REFERENCES

- X00RE1505730 Is. 1.0 "Design Values for Nomex Over Expanded Honeycomb Cores to AIMS 11-01-005" Airbus Technical Report

GRADE A:

PHYSICAL PROPERTIES

Cell size	4.8 mm
Density	29 kg/m ³
Max. Operational Limit (MOL)	135 °C

CTE (1/°C)	Range	-55..20°C	20..70°C	70..135°C
	α	-	-	-

ELASTIC MODULI

Property	Mean Values (MPa)				B-basis K_B	<i>E-KDF</i>		B-basis Values (MPa)			
	-55°C/Dry	RT/Dry	70°C/W	135°C/W		70°C/W	135°C/W	-55°C/Dry	RT/Dry	70°C/W	135°C/W
E_{33c}	31.5	31.5	22.1	16.1	0.74	0.702	0.511	23.3	23.3	16.4	11.9
G_{13}	13.9	13.9	9.9	8.7	0.74	0.713	0.626	10.3	10.3	7.3	6.4
G_{23}	24.1	24.1	18.8	14.9	0.74	0.780	0.618	17.8	17.8	13.9	11.1

STRENGTH

Property	Mean Values (MPa)				B-basis K_B	<i>E-KDF</i>		B-basis Values (MPa)			
	-55°C/Dry	RT/Dry	70°C/W	135°C/W		70°C/W	135°C/W	-55°C/Dry	RT/Dry	70°C/W	135°C/W
F_{33c}	0.65	0.65	0.46	0.31	0.74	0.707	0.477	0.45	0.45	0.34	0.23
F_{13}	0.39	0.39	0.27	0.25	0.74	0.692	0.641	0.25	0.25	0.20	0.19
F_{23}	0.41	0.41	0.27	0.27	0.74	0.658	0.658	0.25	0.25	0.20	0.20

OTHER PROPERTIES

None.

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GRADE F:

PHYSICAL PROPERTIES

Cell size	4.8 mm
Density	48 kg/m ³
Max. Operational Limit (MOL)	135 °C

CTE (1/°C)	Range	-55..20°C	20..70°C	70..135°C
	α	-	-	-

ELASTIC MODULI

Property	Mean Values (MPa)				B-basis K_B	<i>E-KDF</i>		B-basis Values (MPa)			
	-55°C/Dry	RT/Dry	70°C/W	135°C/W		70°C/W	135°C/W	-55°C/Dry	RT/Dry	70°C/W	135°C/W
E_{33c}	89.5	89.5	62.9	45.8	0.74	0.702	0.511	64.9	64.9	45.6	33.9
G_{13}	19.1	19.1	14.3	12.4	0.74	0.744	0.645	14.1	14.1	10.5	9.1
G_{23}	44.7	44.7	40.7	35.5	0.74	0.909	0.782	33.1	33.1	30.1	25.9

STRENGTH

Property	Mean Values (MPa)				B-basis K_B	<i>E-KDF</i>		B-basis Values (MPa)			
	-55°C/Dry	RT/Dry	70°C/W	135°C/W		70°C/W	135°C/W	-55°C/Dry	RT/Dry	70°C/W	135°C/W
F_{33c}	2.35	2.35	1.67	1.24	0.74	0.710	0.527	1.55	1.55	1.24	0.92
F_{13}	0.71	0.71	0.54	0.52	0.74	0.760	0.732	0.52	0.52	0.40	0.39
F_{23}	0.88	0.88	0.56	0.52	0.74	0.636	0.591	0.52	0.52	0.42	0.39

OTHER PROPERTIES

None.

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GRADE H:

PHYSICAL PROPERTIES

Cell size	4.8 mm
Density	64 kg/m ³
Max. Operational Limit (MOL)	135 °C

CTE (1/°C)	Range	-55..20°C	20..70°C	70..135°C
	α	-	-	-

ELASTIC MODULI

Property	Mean Values (MPa)				B-basis K_B	<i>E-KDF</i>		B-basis Values (MPa)			
	-55°C/Dry	RT/Dry	70°C/W	135°C/W		70°C/W	135°C/W	-55°C/Dry	RT/Dry	70°C/W	135°C/W
E_{33c}	138	138	96.9	70.6	0.74	0.702	0.511	102.2	102.2	71.8	52.3
G_{13}	21.4	21.4	17.6	17.6	0.74	0.817	0.817	15.9	15.9	13.0	13.0
G_{23}	57.9	57.9	51.6	43.3	0.74	0.891	0.747	42.8	42.8	38.2	32.1

STRENGTH

Property	Mean Values (MPa)				B-basis K_B	<i>E-KDF</i>		B-basis Values (MPa)			
	-55°C/Dry	RT/Dry	70°C/W	135°C/W		70°C/W	135°C/W	-55°C/Dry	RT/Dry	70°C/W	135°C/W
F_{33c}	4.07	4.07	2.48	1.94	0.74	0.609	0.476	2.16	2.16	1.84	1.44
F_{13}	1.03	1.03	0.62	0.50	0.74	0.602	0.485	0.67	0.67	0.46	0.37
F_{23}	1.13	1.13	0.80	0.80	0.74	0.708	0.708	0.74	0.74	0.59	0.59

OTHER PROPERTIES

None.